

Canonical Linear Core, Generated Hull, and Duality for Non-Even Orlicz Cones

Candidate major partial result for arXiv:2204.12282

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Status. Candidate major partial, likely valid; human review requested. The broad program of a complete nonseparable Orlicz-cone theory and a systematic calculus directly on locally convex cones remains open.

1 Source question and scope

The source is Thomas Ruf, *Non-Separably Valued Orlicz Spaces I*, arXiv:2204.12282v2 (2023) [1]. On PDF page 3 the paper asks:

Can an analogue theory be developed for Orlicz cones generated by a non-even convex Orlicz integrand $\phi: \Omega \times X \rightarrow [0, \infty]$?

It then asks for an associated subdifferential calculus on locally convex cones, motivated by unilateral constraints. Figure 1 is a crop of the source page.

NON-SEPARABLY VALUED ORLICZ SPACES I

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a subsequent part of the present paper.

Open and follow-up questions: We assume the Orlicz integrand φ to vanish continuously at the origin. However, the Orlicz space remains a Banach space if this assumption is dropped. What then is the dual space of $L_\varphi(\mu)$? At least for $\varphi(\omega, x) = \varphi(x)$ we know that a discontinuity at the origin can always be avoided by adapting the range space, but in general we cannot say. The assumption of an even Orlicz integrand restricts. Can an analogue theory be developed for Orlicz cones generated by a non-even convex Orlicz integrand $\phi: \Omega \times X \rightarrow [0, \infty]$? Can one devise an associated subdifferential calculus for functionals on locally convex cones that, together with the Orlicz cones $L_\phi(\mu)$, provides a systematic, more powerful approach to treating unilateral constraints such as obstacle conditions on spaces of smooth functions or other strong asymmetries of the integrand?

Figure 1: The open and follow-up questions in arXiv:2204.12282v2, p. 3.

This packet gives a canonical symmetric core, a canonical generated linear hull valid without separability, exact cone-dual extension and conjugate formulas, an attained-subdifferential rule, and a pointwise Banach-Orlicz realization under finite-dimensional normal-integrand hypotheses. It does not claim the full program.

2 Setting

Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, let X be a real Banach space, and let $\mathcal{E}$ be the vector space of a.e. classes of strongly measurable maps $\Omega \rightarrow X$. Assume that

$$\phi: \Omega \times X \longrightarrow [0, \infty]$$

is integrally separably measurable in the sense of [1], and that for a.e. ω the function $\phi_\omega = \phi(\omega, \cdot)$ is proper, lower semicontinuous and convex, satisfies $\phi_\omega(0) = 0$, is continuous at 0, and has bounded sublevel sets. We do *not* assume evenness. Define

$$I(u) := \int_{\Omega} \phi(\omega, u(\omega)) d\mu, \quad p(u) := \inf\{\alpha > 0 : I(u/\alpha) \leq 1\},$$

with the usual extended-value conventions, and put

$$\mathcal{C} := \{u \in \mathcal{E} : p(u) < \infty\}.$$

Convexity at the origin makes p an extended sublinear gauge, so $\mathcal{C}$ is a convex cone. The elementary modular inequalities used below are

$$p(z) \leq 1 \implies I(z) \leq p(z), \quad I(z) \leq M < \infty \implies p(z) \leq \max\{1, M\}. \quad (2.1)$$

The first follows by radial left-continuity; the second follows by convex rescaling.

3 Idea of the proof

There are two different symmetric objects hidden in a one-sided Orlicz cone. To stay *inside* the cone while allowing signs, a vector and its negative must both be admissible. This forces the reflected maximum

$$\phi_{\text{bi}}(\omega, x) = \max\{\phi(\omega, x), \phi(\omega, -x)\}.$$

It yields the largest linear subspace contained in the cone.

To generate a linear space from the *whole* cone, one instead writes $u = v - w$ with $v, w \in \mathcal{C}$. Pointwise minimization over such decompositions can fail to commute with integration in a nonseparable range. The key selection-free replacement is therefore the function-space infimal convolution

$$J(u) = \inf_{v \in \mathcal{E}} \{I(v) + I(v - u)\}.$$

Its Luxemburg domain is exactly $\mathcal{C} - \mathcal{C}$. Its symmetric gauge is equivalent to the Grothendieck quotient norm obtained by minimizing $p(v) + p(w)$ over $u = v - w$. This quotient description makes the dual extension theorem automatic, while expanding the Fenchel conjugate separates the forward and backward terms exactly. When finite-dimensional measurable selection is available, J becomes the integral of the pointwise infimal convolution and the hull becomes an ordinary Banach Orlicz space.

4 The maximal linear core

Theorem 1 (canonical maximal core). *Set*

$$\phi_{\text{bi}}(\omega, x) := \max\{\phi(\omega, x), \phi(\omega, -x)\}.$$

Then ϕ_{bi} is an even Orlicz integrand and

$$\mathcal{C} \cap (-\mathcal{C}) = L_{\phi_{\text{bi}}}(\mu). \quad (4.1)$$

This is the unique maximal vector subspace contained in $\mathcal{C}$. On this space, if p_{bi} is the Luxemburg norm generated by ϕ_{bi} , then

$$\max\{p(u), p(-u)\} \leq p_{\text{bi}}(u) \leq 2 \max\{p(u), p(-u)\}. \quad (4.2)$$

The universal constant 2 cannot be decreased.

Proof. The maximum of two proper lower semicontinuous convex functions is again lower semicontinuous and convex. The definition makes ϕ_{bi} even; continuity at zero, coercivity, and integral separable measurability pass to the maximum. Thus the even theory of [1] applies.

Since $\phi_{\text{bi}} \geq \phi$ and $\phi_{\text{bi}}(\omega, x) \geq \phi(\omega, -x)$,

$$p(u) \leq p_{\text{bi}}(u), \quad p(-u) \leq p_{\text{bi}}(u).$$

This proves the left inequality in (4.2) and the inclusion $L_{\phi_{\text{bi}}} \subset \mathcal{C} \cap (-\mathcal{C})$.

Conversely, let $u, -u \in \mathcal{C}$ and take $a > \max\{p(u), p(-u)\}$. Then $I(u/a) \leq 1$ and $I(-u/a) \leq 1$. Hence

$$I_{\text{bi}}(u/a) \leq I(u/a) + I(-u/a) \leq 2.$$

Convexity at zero gives $I_{\text{bi}}(u/(2a)) \leq 1$, so $p_{\text{bi}}(u) \leq 2a$. Letting a decrease proves the right inequality and the reverse inclusion in (4.1).

The intersection $\mathcal{C} \cap (-\mathcal{C})$ is a vector space by (4.1). If V is any vector subspace contained in $\mathcal{C}$, then $u \in V$ implies $-u \in V \subset \mathcal{C}$, so $V \subset \mathcal{C} \cap (-\mathcal{C})$. Optimality of 2 is proved in Example 7. $\square$

Remark 2. The norm $u \mapsto \max\{p(u), p(-u)\}$ is therefore complete on the core, because it is equivalent to the Banach-Orlicz norm p_{bi} .

5 The generated hull and its Grothendieck norm

Define the even convex functional

$$J(u) := \inf_{v \in \mathcal{E}} \{I(v) + I(v - u)\}, \quad p_J(u) := \inf\{\alpha > 0 : J(u/\alpha) \leq 1\}. \quad (5.1)$$

Also put

$$\mathcal{H} := \mathcal{C} - \mathcal{C}, \quad r(u) := \inf_{\substack{v, w \in \mathcal{C} \\ u = v - w}} \{p(v) + p(w)\}. \quad (5.2)$$

Theorem 3 (selection-free hull and sharp norm comparison). *The functional J is even and convex, and*

$$\{u \in \mathcal{E} : p_J(u) < \infty\} = \mathcal{H} = \mathcal{C} - \mathcal{C}. \quad (5.3)$$

Moreover, p_J and r are norms on $\mathcal{H}$ and

$$p_J(u) \leq r(u) \leq 2p_J(u) \quad (u \in \mathcal{H}). \quad (5.4)$$

The constant 2 cannot be decreased, even for an even autonomous integrand on a one-point measure space.

Proof. Writing $w = v - u$, formula (5.1) is the infimal convolution of I with its reflection. It is convex, and

$$J(-u) = \inf_v \{I(v) + I(v + u)\} = \inf_w \{I(w - u) + I(w)\} = J(u).$$

If $u = v - w$ with $v, w \in \mathcal{C}$, choose a common scale $a > 0$ at which $I(v/a)$ and $I(w/a)$ are finite. Then $J(u/a) \leq I(v/a) + I(w/a) < \infty$, and a further convex rescaling makes this quantity at most 1. Thus $p_J(u) < \infty$. Conversely, if $p_J(u) < \infty$, then $J(u/a) < \infty$ for some $a > 0$. There is a $v \in \mathcal{E}$ for which both $I(v)$ and $I(v - u/a)$ are finite. Putting $w = v - u/a$ gives $u = av - aw$ with $av, aw \in \mathcal{C}$. This proves (5.3) without an attainment or measurable-selection assumption.

For the first inequality in (5.4), fix $u = v - w$, take $a > p(v)$ and $b > p(w)$, and put $s = a + b$. Convexity gives

$$I(v/s) \leq \frac{a}{s}I(v/a) \leq \frac{a}{s}, \quad I(w/s) \leq \frac{b}{s}I(w/b) \leq \frac{b}{s}.$$

Therefore $J(u/s) \leq 1$ and $p_J(u) \leq a + b$. Take the infimum.

For the reverse inequality, take $a > p_J(u)$, so $J(u/a) \leq 1$ after decreasing an admissible scale if necessary. For every $\varepsilon > 0$ there are $v, w \in \mathcal{E}$ with $v - w = u/a$ and

$$I(v) + I(w) \leq 1 + \varepsilon.$$

Each term is at most $1 + \varepsilon$, hence convex rescaling gives $p(v), p(w) \leq 1 + \varepsilon$. Consequently $r(u) \leq 2a(1 + \varepsilon)$. Let $\varepsilon \downarrow 0$ and $a \downarrow p_J(u)$.

It remains only to note separation. If $r(u) = 0$, choose $u = v_n - w_n$ with $p(v_n) + p(w_n) \leq 2^{-2n}$. By (2.1),

$$I(2^n v_n) + I(2^n w_n) \leq 2^{1-n}.$$

The sum of these integrals is finite. Hence both integrands tend to zero a.e. Coercivity then forces $v_n \rightarrow 0$ and $w_n \rightarrow 0$ a.e.; otherwise $2^n \|v_n(\omega)\|$ or $2^n \|w_n(\omega)\|$ would diverge along a subsequence while the corresponding integrand tends to zero. Thus $u = v_n - w_n = 0$ a.e. The equivalence (5.4) separates p_J as well. Optimality is in Example 7. $\square$

6 Cone duality and subdifferential splitting

Call a map $A: \mathcal{C} \rightarrow \mathbb{R}$ a cone-linear functional if it is additive and positively homogeneous. Give it the norm

$$\|A\|_{\mathcal{C}} := \sup\{|A(v)| : v \in \mathcal{C}, p(v) \leq 1\}.$$

Theorem 4 (Grothendieck extension and exact bidirectional duality). (a) *Every cone-linear A with $\|A\|_{\mathcal{C}} < \infty$ has the unique linear extension*

$$\ell(v - w) = A(v) - A(w) \quad (v, w \in \mathcal{C}) \tag{6.1}$$

to $\mathcal{H}$. It is continuous for r , and $\|\ell\|_{r,} = \|A\|_{\mathcal{C}}$.*

(b) *For a linear functional ℓ on $\mathcal{H}$, define*

$$I^*(\ell) := \sup_{v \in \mathcal{C}} \{\ell(v) - I(v)\}, \quad J^*(\ell) := \sup_{u \in \mathcal{H}} \{\ell(u) - J(u)\}.$$

Then, with extended nonnegative values,

$$J^*(\ell) = I^*(\ell) + I^*(-\ell). \tag{6.2}$$

- (c) Suppose $u = v - w$, $v, w \in \mathcal{C}$, attains $J(u) = I(v) + I(w)$. With subdifferentials taken in any fixed separating dual pair containing ℓ ,

$$\ell \in \partial J(u) \iff \ell \in \partial I(v) \text{ and } -\ell \in \partial I(w). \quad (6.3)$$

Proof. If $v - w = v' - w'$, then $v + w' = v' + w$. Additivity gives $A(v) - A(w) = A(v') - A(w')$, so (6.1) is well-defined and plainly linear. For every decomposition,

$$|\ell(v - w)| \leq \|A\|_{\mathcal{C}} \{p(v) + p(w)\}.$$

Thus $\|\ell\|_{r,*} \leq \|A\|_{\mathcal{C}}$. Conversely $r(v) \leq p(v)$ for $v \in \mathcal{C}$, so restriction gives the reverse norm inequality and uniqueness.

For the conjugate, parameterize $u = v - w$:

$$\begin{aligned} J^*(\ell) &= \sup_{u \in \mathcal{H}, v \in \mathcal{E}} \{\ell(u) - I(v) - I(v - u)\} \\ &= \sup_{v, w \in \mathcal{C}} \{\ell(v) - I(v) - \ell(w) - I(w)\} \\ &= I^*(\ell) + I^*(-\ell). \end{aligned}$$

No minimax interchange or attainment is used.

For the last assertion, if the two right-hand subgradient inclusions hold, add their subgradient inequalities for any $u' = v' - w'$ and then take the infimum over decompositions of u' . This gives $J(u') \geq J(u) + \ell(u' - u)$. Conversely, Fenchel equality for $\ell \in \partial J(u)$, the attained decomposition, and (6.2) says

$$[I(v) + I^*(\ell) - \ell(v)] + [I(w) + I^*(-\ell) + \ell(w)] = 0.$$

Both brackets are nonnegative Fenchel gaps, so both vanish. This is exactly the pair of subgradient inclusions in (6.3). $\square$

Corollary 5 (polar formula). *Let*

$$\mathfrak{a}_I(\ell) := \inf_{\alpha > 0} \alpha \{1 + I^*(\ell/\alpha)\}.$$

Then

$$\|\ell\|_{r,*} = \sup_{p(v) \leq 1} |\ell(v)| = \max\{\mathfrak{a}_I(\ell), \mathfrak{a}_I(-\ell)\}. \quad (6.4)$$

Proof. The r -unit ball is the absolutely convex hull of the one-sided unit ball $\{p \leq 1\}$, in the sense of Minkowski gauges. Its support function is therefore $\sup_{p(v) \leq 1} |\ell(v)|$. The one-sided Luxemburg–Amemiya polar identity

$$\sup_{p(v) \leq 1} \ell(v) = \inf_{\alpha > 0} \alpha \{1 + I^*(\ell/\alpha)\}$$

is the usual gauge-conjugacy calculation; the proof in [1, §3.1] uses only convexity, $I(0) = 0$, and radial left-continuity for this one-sided statement. Apply it to ℓ and $-\ell$. $\square$

7 A pointwise Banach-Orlicz hull

Theorem 6 (finite-dimensional pointwise realization). *Assume additionally that $(\Omega, \mathcal{A}, \mu)$ is complete and σ -finite, X is finite-dimensional, and ϕ is a normal integrand. Define*

$$\psi(\omega, x) := \min_{y \in X} \{\phi(\omega, y) + \phi(\omega, y - x)\}. \quad (7.1)$$

Then ψ is an even Orlicz integrand,

$$J(u) = I_\psi(u) \quad (u \in \mathcal{E}), \quad \mathcal{C} - \mathcal{C} = L_\psi(\mu), \quad p_J = \|\cdot\|_\psi. \quad (7.2)$$

In particular, $(\mathcal{C} - \mathcal{C}, p_J)$ is a Banach space and the ordinary even Orlicz theory applies to the generated hull.

Proof. For fixed (ω, x) , coercivity makes $y \mapsto \phi(\omega, y) + \phi(\omega, y - x)$ inf-compact in finite dimensions, so the minimum is attained. Standard inf-projection and measurable-selection results for normal integrands [3, Chapter 14] show that ψ is a normal integrand and provide measurable ε -minimizers.

The infimal convolution of a convex function with its reflection is convex and even. It vanishes continuously at zero because

$$0 \leq \psi(\omega, x) \leq \min\{\phi(\omega, x), \phi(\omega, -x)\} \longrightarrow 0.$$

If $\|x_n\| \rightarrow \infty$ while $\psi(\omega, x_n)$ stayed bounded, minimizing y_n would make both $\phi(\omega, y_n)$ and $\phi(\omega, y_n - x_n)$ bounded. Bounded sublevels would make both sequences bounded, contradicting $x_n = y_n - (y_n - x_n)$. Thus ψ is an even Orlicz integrand.

Every measurable v gives the pointwise upper candidate $\psi(\omega, u(\omega)) \leq \phi(\omega, v(\omega)) + \phi(\omega, v(\omega) - u(\omega))$, hence $I_\psi(u) \leq J(u)$. Measurable ε -minimizers, with an integrable positive error weight on the σ -finite space, give the reverse inequality after integration and $\varepsilon \downarrow 0$. Thus $J = I_\psi$. Now (5.3) gives the set identity, and the gauge identity is immediate. Completeness follows from the even Orlicz-space theorem of [1]. $\square$

8 Sharpness examples

Example 7 (both constants are optimal). **Core.** Let Ω have two atoms of mass $1/2$, let $X = \mathbb{R}$, and for $0 < \varepsilon < 1$ put

$$\phi_\varepsilon(t) = t_+ + \varepsilon t_-, \quad u = (1, -1).$$

The modular is positively homogeneous, so

$$\max\{p(u), p(-u)\} = \frac{1 + \varepsilon}{2}, \quad p_{\text{bi}}(u) = 1.$$

The ratio is $2/(1 + \varepsilon) \rightarrow 2$.

Hull. On a one-point space let

$$\phi_M(t) = M(|t| - 1)_+, \quad M > 0.$$

Then

$$p(t) = \frac{M|t|}{M+1}, \quad J(t) = M(|t| - 2)_+, \quad p_J(t) = \frac{M|t|}{2M+1}.$$

Here p is already symmetric, so $r = p$. Hence

$$\frac{r(t)}{p_J(t)} = \frac{2M+1}{M+1} \longrightarrow 2.$$

9 Verification and novelty bounds

The accompanying script `code/verify_discrete_models.py` checks 500 random finite-atom power models for (4.2), 500 random scalar power models for the coefficient form of (6.2), and the two sharpness families above. These checks are sanity tests, not proof.

The bounded novelty search covered the run’s cheap indexes and exact/close arXiv searches for non-even Orlicz integrands, Orlicz cones, asymmetric Orlicz or modular spaces, symmetrization, and infimal convolution. Ruf’s 2024 thesis repeats the questions [2]. A 2025 paper develops max-symmetrization for abstract quasi-modular pseudometrics [4], but the checked text does not contain the maximal Orlicz-cone core, the generated hull, the Grothendieck dual extension, or the subdifferential split proved here. Novelty confidence is therefore *moderate*, not definitive: terminology in asymmetric modular theory is fragmented.

10 Limitations and review recommendation

- The result is a major partial answer, not a complete construction of the source’s requested theory on locally convex cones.
- In the arbitrary nonseparable case, J is deliberately defined at the function-space level. A pointwise integrand formula would require an inf-projection and measurable-selection theorem that is unavailable under the stated general hypotheses.
- The subdifferential equivalence (6.3) requires attainment of the optimal decomposition. Without attainment, only approximate variants are immediate.
- Human review should focus on the extended-gauge conventions in (5.3), the one-sided polar identity in Corollary 5, and the normal-integrand selection step in Theorem 6.

References

- [1] T. Ruf, *Non-Separably Valued Orlicz Spaces I*, arXiv:2204.12282v2, 2023.
- [2] T. D. Ruf, *Orlicz-type Function Spaces and Generalized Gradient Flows with Degenerate Dissipation Potentials in Non-Reflexive Banach Spaces: Theory and Application*, doctoral dissertation, arXiv:2401.17440, 2024.
- [3] R. T. Rockafellar and R. J.-B. Wets, *Variational Analysis*, Grundlehren der mathematischen Wissenschaften 317, Springer, 1998.
- [4] P. R. Majozi, *On Quasi-Modular Pseudometric Spaces and Asymmetric Uniformities*, arXiv:2510.25200, 2025 (source version checked in 2026).